# Xiaochuan Li


| | | | |
|---|---|---|---|
| Affiliation: | M.Sc. student<br>Human and Artificial Intelligence<br>University of Technology Nuremberg | Homepage: | xchuan-li.github.io |
| | | GitHub: | github.com/xchuan-li |
| E-mail: | xiaochuan.li@utn.de | | |


How do people and artificial systems understand the concepts, evidence, and norms in language, and turn them into inferences and choices? Linguistics and cognitive science provide precise questions about meaning and decades of carefully measured human behaviour. Taken over to language models, those questions become controlled experiments: tests of behaviour show what a model does, and interpretability methods show where in the model it happens.

## Education

Oct. 2025 – exp. 2027 **University of Technology Nuremberg**, Germany
*M.Sc. Human and Artificial Intelligence*
Designing and analysing behavioural experiments; cognitive psychology; training neural networks; interpretability methods for language models.

Oct. 2024 – Jul. 2025 **University of Marburg**, Germany
*German for graduate study*

Sep. 2020 – Jun. 2024 **Shenzhen University**, China
*B.A. Philosophy*
Formal logic and proof, including modal logic and Gödel's incompleteness theorems; philosophy of language; psychology of reading.

## Research

Apr. 2026 – Mar. 2027 **Moral framing and dilemma choices in language models**
*Supervised at the German Research Center for Artificial Intelligence (DFKI)*
Investigated how moral wording affects dilemma choices. Designed and ran controlled experiments and activation interventions in 7B and 14B models for the utilitarian-framing branch, with checks of answer labels, option positions, and task understanding; manuscript in preparation.

Sep. 2026 – **Property updating in modified nouns in language models**
*Independent project*
Contrasted properties that depend on an added feature with properties that only co-occur with it (13 feature groups, 477 items, three instruction-tuned models), using predictors fixed outside the models: a dependency table, human feature norms and word-association norms; exploratory activation patching. Manuscript in preparation.

Sep. 2026 – **Modal language in preschool children**
*M.Sc. thesis, design stage · part of a PhD project at the Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig*
Asks how preschool children come to distinguish possibility from necessity in language, and how this relates to their nonlinguistic reasoning about possibilities; a behavioural study; data collection has not started. It lays the groundwork for later work on modal language in language models.

## Projects

| | |
|---|---|
| Sep. 2026 | Typicality in referent choice: what the description leaves open<br>*Independent theoretical study.* Distinguishes three conditions for typicality in referent choice and the limits of what a forced-choice task can establish. Li, X. (2026). Working paper. lingbuzz/010343 · doi:10.5281/zenodo.22855036 |
| May – Aug. 2026 | Cross-lingual ProtoBias<br>*Single-author course project.* Evaluated two VLM families on fixed image pairs across seven prompt languages (about 12,600 judgments); single-author report with code. report · code |
| May – Jul. 2026 | Street-view country classification<br>*Team course project, Deep Learning.* My part: CNN training, ablations, and performance analysis. |
| Nov. 2025 – Feb. 2026 | Fact-checking after ChatGPT exposure<br>*Course study, Foundations in Psychology and Empirical Study Design.* Between-subjects experiment on optional checking after ChatGPT or Wikipedia answers (49 participants, 47 analysed); co-designed the study, ran the data analysis and wrote the report. |
| Dec. 2023 – May 2024 | Leibniz's logical system and characteristica universalis<br>*Undergraduate thesis · Shenzhen University.* Analysed Leibniz's metaphysics, logic and proposal for a universal symbolic language. |

## Employment

| | |
|---|---|
| Apr. 2026 – present | **University of Technology Nuremberg**, Germany<br>*IT Service Desk (part-time)*<br>Delivered student IT onboarding and individual support; handled system installation, service reporting, and inventory. |
| Sep. – Dec. 2021 | **Shenzhen Learning Disorders Association**, China<br>*Internship · Learning assessment at Yucai No. 2 Primary School*<br>Prepared character-reading materials, administered timed assessments, and analysed response data to support the assessment of learning difficulties; contributed to teaching items and individual learning plans. |

## Skills and languages

| | |
|---|---|
| Language models | Controlled behavioural evaluation, minimal pairs; activation patching, low-rank analysis, probing, steering |
| Human data | Experimental design, PsychoPy/Pavlovia, JASP; testing children |
| Programming | Python (PyTorch, Hugging Face Transformers, pandas, scikit-learn), Git, LaTeX, Slurm on an HPC cluster |
| Languages | Mandarin Chinese (native), German (C1), English (C1), Japanese (learning) |